



MIMO Systems in Wireless Communications

HW- 1

Master's Educational Program: IoT and Wireless Communication

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Results and Discussion:

1.1 Singular Value Distribution Across Channel Files:

The averaged singular values reveal clear differences in spatial structure between the considered channel realizations.

From the computed results:

CHANNEL	Σ_2/Σ_1	GAP (DB)	INTERPRETATION
LINK_CHAN_1	0.652	3.72 dB	Moderate spatial richness
LINK_CHAN_2	0.515	5.76 dB	Strong dominant mode
LINK_CHAN_3	0.647	3.78 dB	Moderate spatial richness
LINK_CHAN_PATH	0.859	1.32 dB	Highly rank-rich
LINK_CHAN_SPEED	0.719	2.86 dB	Good multiplexing potential

Table 1.1

Key Observations:

link_chan_2 – Dominant First Mode

This channel shows the largest singular value gap (5.76 dB) and the lowest σ_2/σ_1 ratio (0.515).

This indicates:

- One dominant spatial direction
- Weaker secondary spatial mode
- Channel behaves close to rank-1

Such channels favor **single-stream beamforming**, because most of the power is concentrated in the first eigenmode.

This behavior is also visible in the normalized singular value plot for link_chan_2, where the first singular value is significantly larger than the others.

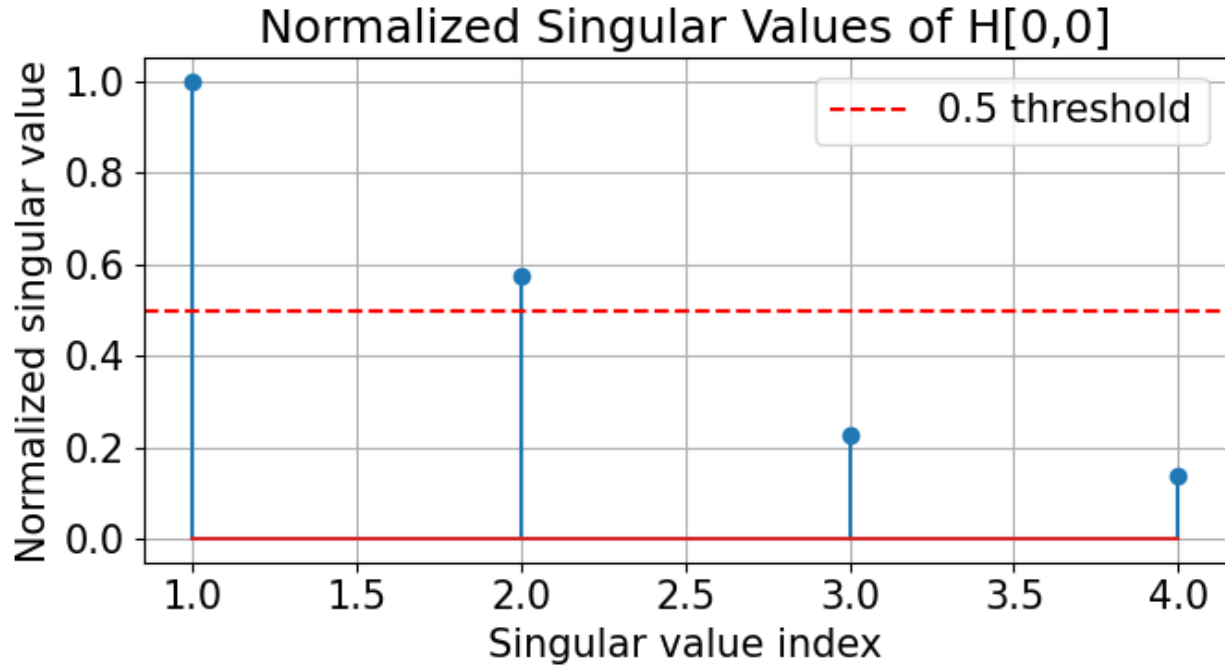


Fig 1

link_chan_PATH – Rich Multipath Environment

This channel has the smallest gap (1.32 dB) and the highest σ_2/σ_1 ratio (0.859).

This indicates:

- Two dominant spatial modes with comparable strength
- Strong multipath richness
- High spatial rank

This channel is highly suitable for **dual-stream transmission**, since both eigenmodes can carry comparable energy.

In the heatmap (Fig 2), link_chan_PATH clearly maintains higher power in the second singular mode compared to other channels.

link_chan_SPEED – Moderate Rank Richness

With $\sigma_2/\sigma_1 \approx 0.72$ and gap ≈ 2.86 dB:

- This channel supports multiplexing reasonably well.
- Not as rich as PATH, but significantly better than link_chan_2.

This suggests that mobility (speed) introduces spatial variation but still preserves multipath richness.

Link_chan_1 and link_chan_3 – Similar Behavior

Both channels show similar σ_2/σ_1 (~ 0.65) and similar gaps (~ 3.7 dB).

These channels:

- Have moderate spatial richness
- Support two streams, but the second stream will be weaker
- Are balanced between beamforming and multiplexing suitability

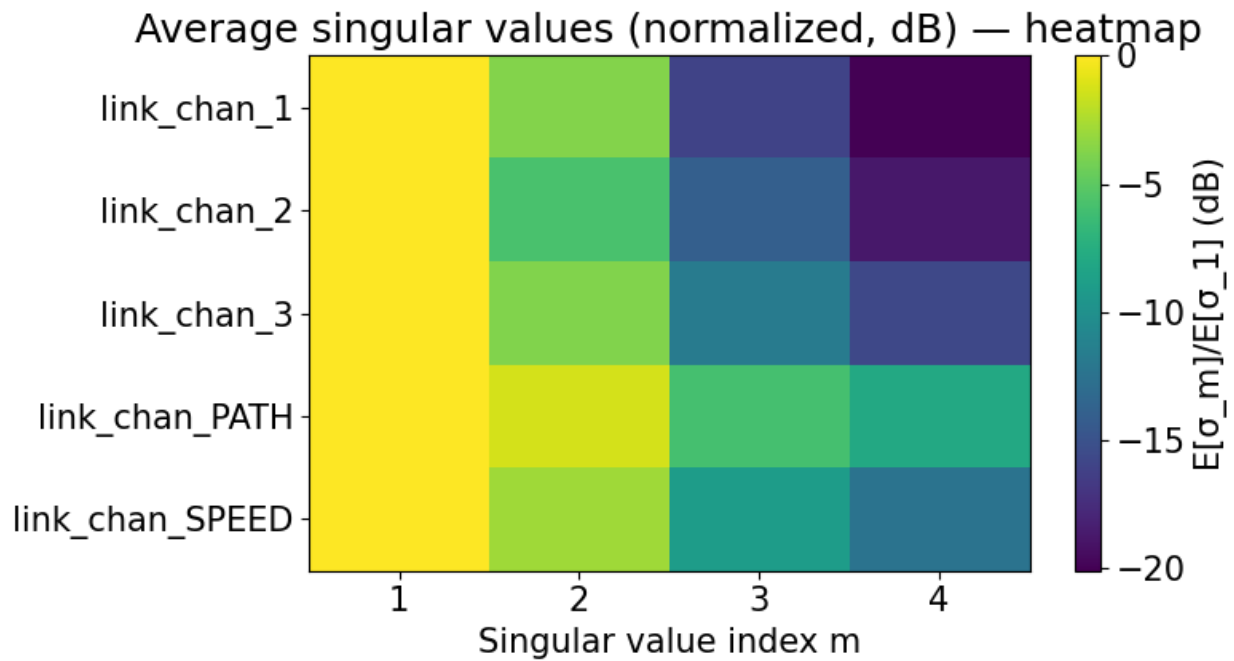


Fig (2)

1.2 Interpretation of link_chan_2 (Detailed Case Study)

Normalized Singular Values

The normalized singular value plot shows (Fig 1):

- First singular value = 1 (reference)
- Second singular value ≈ 0.58
- Third and fourth drop significantly

This confirms that:

- Channel energy is strongly concentrated in the first spatial mode.
- Higher modes contribute much less.

Heatmap Analysis

The heatmap further confirms (Fig 2):

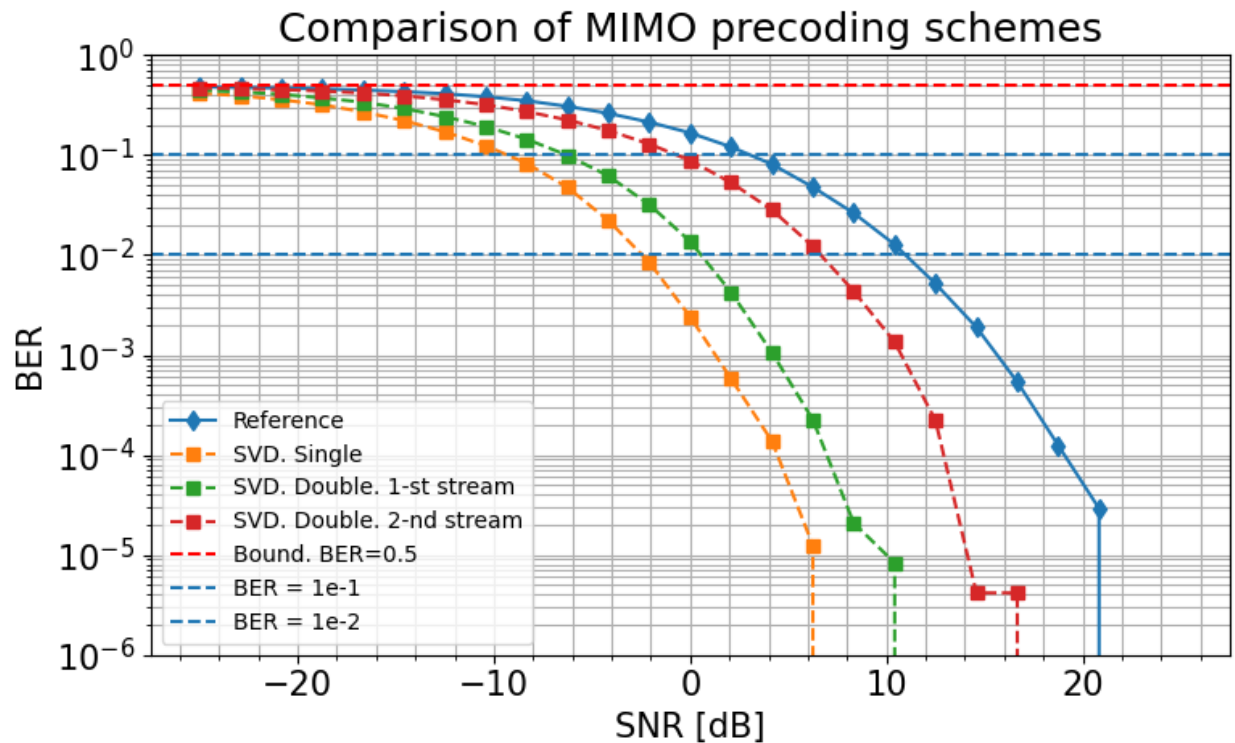
- link_chan_2 exhibits rapid decay across singular value indices.
- Compared to PATH and SPEED, the third and fourth modes are much weaker.

This spatial structure explains why:

- Single-stream transmission performs very well.
- Dual-stream performance will be limited by the weaker second mode.

1.3 BER Performance Analysis (link_chan_2)

The BER plot for link_chan_2 shows(Fig 3):



○ Fig 3

- SVD single stream significantly outperforms reference precoding.
- Horizontal SNR offset:
 - 12.37 dB at BER = 10^{-1}**
 - 13.42 dB at BER = 10^{-2}**

```

... BER=1e-01
    Reference SNR: 3.06 dB
    SVD SNR:      -9.31 dB
    Horizontal offset: 12.37 dB

BER=1e-02
    Reference SNR: 10.97 dB
    SVD SNR:      -2.45 dB
    Horizontal offset: 13.42 dB
  
```

The BER curves clearly demonstrate the significant performance gain achieved by SVD-based precoding compared to uniform transmission. In the single-stream case, the SVD curve is

shifted substantially to the left relative to the reference (uniform) curve. This indicates that the same BER can be achieved at a much lower SNR. For example, at $\text{BER} = 10^{-2}$, SVD requires approximately 12 dB less SNR than uniform precoding.

This improvement occurs because SVD exploits channel knowledge and transmits along the strongest singular vector of the channel matrix. In other words, all transmit power is concentrated in the dominant spatial mode corresponding to the largest singular value. This maximizes the effective channel gain and therefore improves reliability.

In contrast, uniform precoding does not adapt to the channel. It distributes transmit power equally across antennas without considering the channel eigenstructure. As a result, part of the transmitted energy is directed into weaker spatial modes, which reduces the effective received signal strength and degrades BER performance.

In the dual-stream case, two independent data symbols are transmitted simultaneously using the two strongest singular directions. However, the total transmit power is divided between the two streams. Since the second singular value is smaller than the first one, and the available power per stream is reduced, the second stream experiences significantly lower effective channel gain. This is clearly visible in the figure, where the second stream exhibits worse BER performance compared to the first stream.

Overall, the results highlight the trade-off between reliability and spatial multiplexing. Single-stream SVD maximizes reliability by focusing all power on the strongest spatial mode, while dual-stream transmission increases data rate but reduces robustness, especially for the weaker spatial mode. The performance gap between the two streams directly reflects the difference between the first and second singular values of the channel.

Thus, `link_chan_2` is more suitable for **single-stream transmission**.

1.4 Explanation of Uniform vs SVD Precoding Performance:

Although uniform precoding distributes the transmit signal equally across all antennas, it does not take the channel structure into account. As a result, the transmitted signal is not aligned with the dominant spatial eigenmode of the channel. Consequently, part of the transmit power is projected onto weaker singular modes, reducing the effective received SNR. In contrast, SVD precoding exploits full channel state information and aligns transmission with the strongest singular vector. This ensures coherent combining at the receiver and maximizes beamforming gain. Therefore, even though both schemes use all antennas, SVD precoding concentrates energy in the most favorable spatial direction, leading to the substantial SNR gain observed in the BER results.

1.5 Single vs Dual Stream Suitability:

The effectiveness of spatial multiplexing depends directly on the singular value distribution.

When $\sigma_2 \approx \sigma_1$ (e.g., link_chan_PATH)

- Two streams have similar strength.
- Dual-stream transmission is efficient.
- Spatial multiplexing provides throughput gain.

When $\sigma_2 \ll \sigma_1$ (e.g., link_chan_2)

- The second stream suffers from low effective channel gain.
- Noise amplification becomes significant.
- Single-stream beamforming is preferable.

Questions to Answer:

1. Why is it not possible to reliably transmit multiple independent data streams using the uniform precoder, in contrast to SVD-based precoding?
2. What is the optimal strategy for normalizing SVD precoding vectors in single-stream and multi-stream transmission scenarios?
3. How to define a maximum possible streams that can be reliably transmitted?

The questions have been answered in the notebook markdown:

2. Interpretation of the Spatial Spectrum

The spatial spectrum shows how the channel energy is distributed over different spatial directions. The two plots present the same spectrum in linear scale (left) and in dB scale (right).

A clear dominant energy concentration can be observed around $u \approx -0.2$ and $v \approx 0$. This indicates that most of the received signal power arrives from one main spatial direction. The energy is strongly focused rather than spread across many directions.

The linear-scale plot highlights the dominant peak, while the dB-scale plot makes weaker components more visible. In the dB representation, secondary sidelobes and weaker paths can be observed at lower power levels, but they are significantly weaker than the main peak. The presence of a single strong peak suggests that the channel is dominated by one major propagation path. This implies that the channel structure is close to rank-1, meaning that most of the signal power is concentrated in one spatial mode.

Such a directional structure explains why single-stream SVD precoding performs very well in the BER results. Since most energy comes from one dominant direction, transmitting along that strongest spatial mode maximizes the received signal power and improves reliability.

Overall, the spatial spectrum confirms that the channel has a strong directional component with limited angular spread.

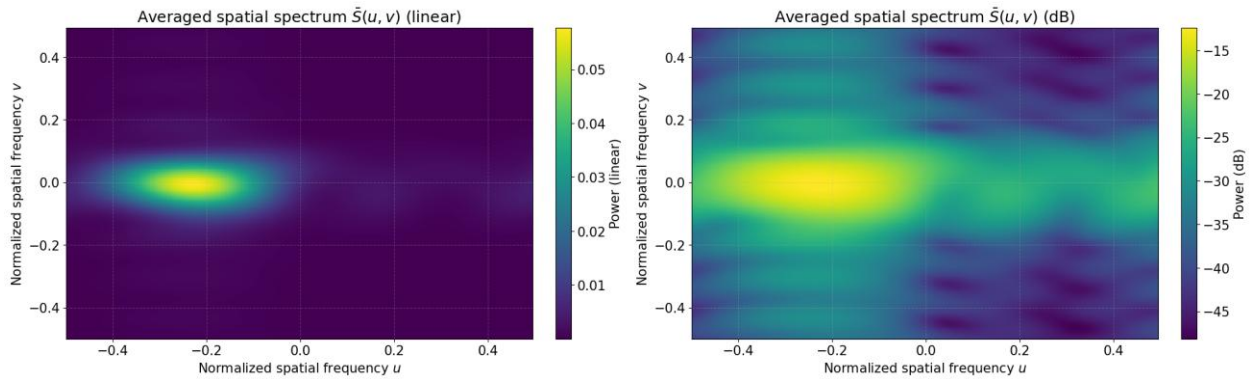


Fig 4

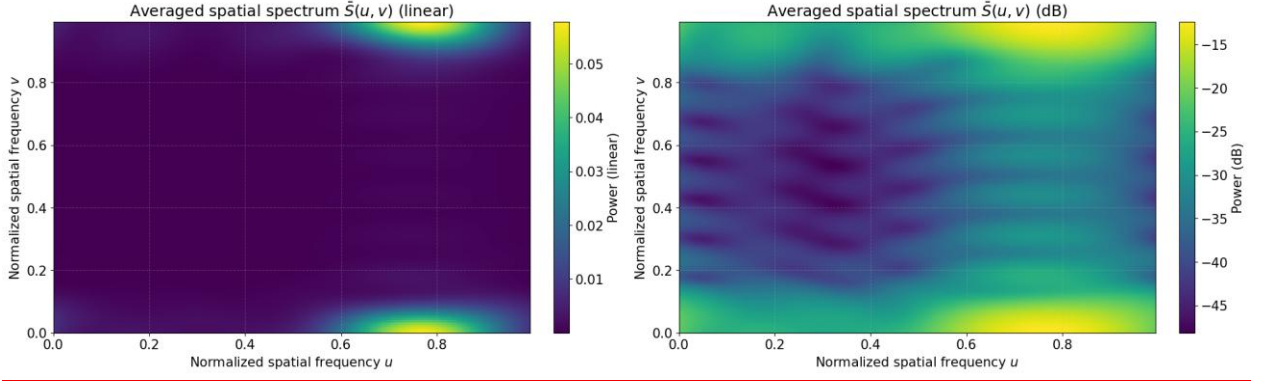


Fig 5

The spatial spectrum was visualized using both the unit grid $u, v \in [0,1)$ and the centered grid $u, v \in [-0.5,0.5)$. Although both representations are mathematically equivalent due to the periodicity of the DFT, their visual appearance differs.

In the centered representation, a single dominant peak is clearly visible around $u \approx 0.8$ and $v \approx 0$. This indicates that the channel energy is concentrated in one main spatial direction, suggesting a near rank-1 channel structure.

In the unit-grid representation, the dominant peak appears partially split across the boundaries of the interval $[0,1)$. This occurs because spatial frequency is periodic, and negative frequencies in the centered grid are mapped to values close to 1 in the unit grid. Therefore, the apparent multiple lobes do not correspond to additional propagation paths, but rather to the wrap-around property of the DFT representation.

2.1 Angular-domain spatial spectrum (ϕ, θ).

To improve physical interpretability, the averaged spatial spectrum was additionally mapped from normalized spatial frequencies (u, v) into azimuth–elevation angles (ϕ, θ). The resulting heatmap shows a clear dominant region concentrated near **moderate positive azimuth (around -30°)** and **near-zero elevation (around 0°)**, indicating that the channel energy is dominated by a main propagation direction. The spectrum is relatively narrow in elevation and wider in azimuth, suggesting stronger angular selectivity vertically and more spread horizontally. Since the mapping was implemented using **nearest-neighbor lookup**, the angular spectrum may exhibit slight blockiness/quantization artifacts; however, the dominant direction remains consistent with the (u, v) spectrum peak.

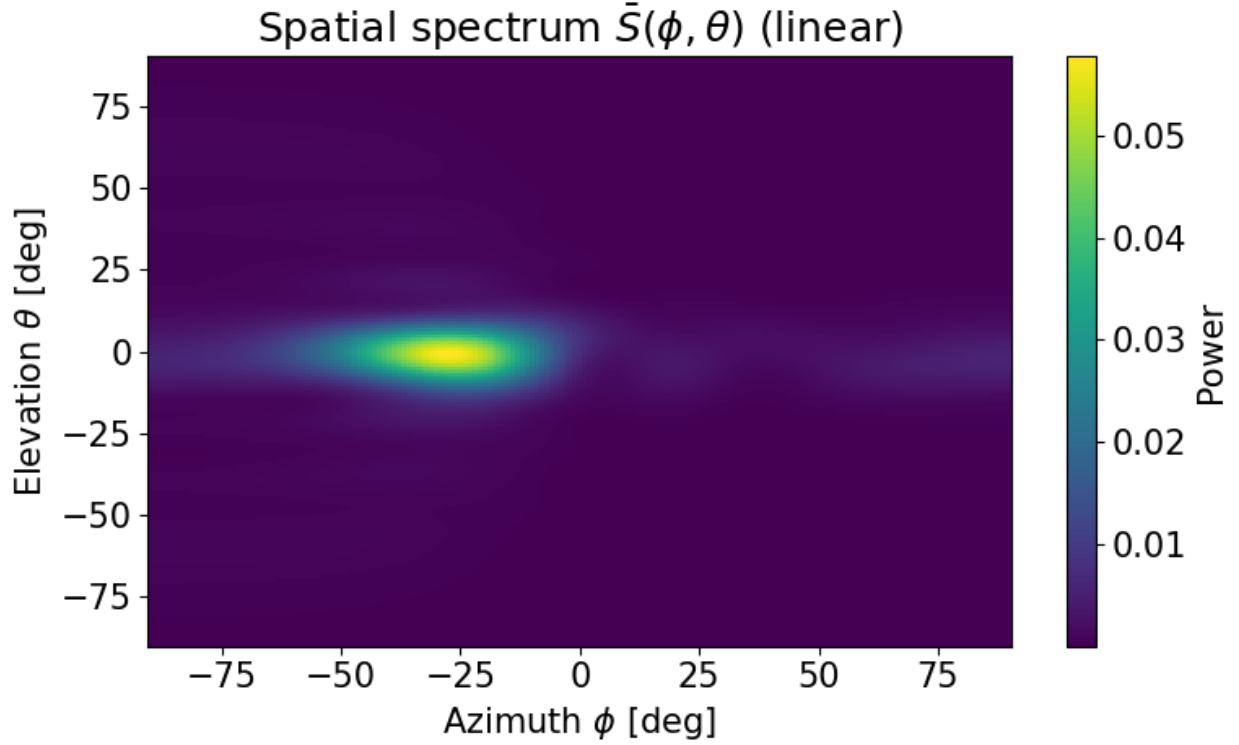


Fig 6

4. Beam-domain channel approximation: implementation summary (FFT/IFFT)

To exploit angular sparsity, the BS antenna-domain channel vector $h \in \mathbb{C}^N$ (with $N = 64$) is transformed into a beam (DFT) domain using an orthonormal FFT:

$$z = Fh$$

$$z = \text{FFT}(h)$$

Only the strongest B beam coefficients (largest magnitudes $|z_i|$) were retained, and all other components were set to zero. Then the approximated channel is reconstructed with the inverse FFT:

$$\hat{h} = F^H \hat{z}$$

$$\hat{h} = \text{IFFT}(\hat{z})$$

This is done per (k, m) (subcarrier and UE antenna), producing an approximated channel $\hat{H}^{(B)}$. The SVD-based precoder is computed using $\hat{H}^{(B)}$, while transmission still occurs through the **true** channel H . Therefore, performance degradation directly reflects a mismatch due to beam truncation.

3.1 BER–SNR results for $B \in \{1, 4, 8\}$: observations

General trend

The BER curves show a clear monotonic improvement as the number of retained beams increases:

$$B = 1 \text{ (worst), } B = 4, B = 8 \text{ (best, closest to full SVD)}$$

This is expected because increasing B retains more dominant angular components, which makes $\hat{H}^{(B)}$ a better approximation of H , and thus produces a more accurate SVD precoder.

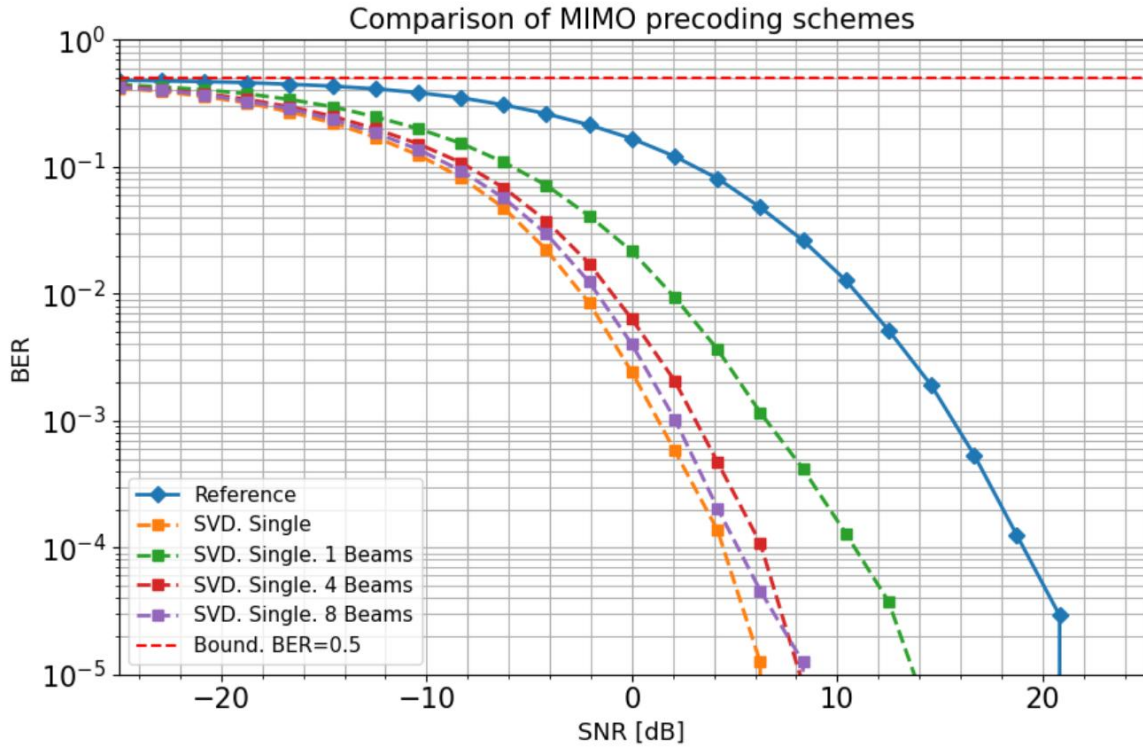


Fig 7

3.2 Comparison to baselines in the plot

My plot includes (Fig 7):

- **Reference:** worst curve (requires much higher SNR for same BER)
- **SVD Single:** best among your curves (full-channel SVD precoding)
- **Beam-approximated SVD:** $B = 1$, $B = 4$, $B = 8$

Key qualitative observation:

- The **full SVD Single** curve achieves low BER at much smaller SNR than the **Reference** curve, confirming that SVD/matched precoding strongly improves effective channel gain.
- The beam-approximated curves lie **between** the Reference and full SVD:
 - $B = 1$ is the farthest from full SVD
 - $B = 8$ is closest to full SVD and nearly overlaps it at moderate/high SNR

Why $B = 1$ performs noticeably worse

The $B = 1$ approximation introduces strong model mismatch because it forces the channel to be rank-1 in the beam domain, so the precoder aligns with only one dominant beam and ignores residual paths that still contribute significantly.

Why $B = 4$ and $B = 8$ improve quickly

When B increases:

- more dominant paths (or clusters of angular energy) are included,
- channel power captured by $\hat{H}^{(B)}$ increases,
- precoder computed from $\hat{H}^{(B)}$ becomes closer to the optimal precoder of H .

Thus, both:

1. **array gain increases** (better alignment with the true channel), and
2. **inter-beam “leakage” is reduced** (less error due to truncation).

3.3 Beam Energy Capture:

The beam energy ratio quantifies how much total channel power is preserved when only the strongest B beams are retained.

Measured results:

- **B = 1:** 46.7% of total energy captured
- **B = 4:** 74.3% of total energy captured
- **B = 8:** 85.4% of total energy captured

1 0.4668828854854472
4 0.7431811579426401
8 0.8540681102042166

Observation

- The channel is **moderately sparse** in the beam domain.
- Nearly half of the total channel energy is concentrated in a single dominant beam.
- However, significant additional energy is distributed across other beams.
- Increasing B from 1 to 4 yields a large gain in captured energy ($\sim +27\%$).
- Increasing B from 4 to 8 yields a smaller but still meaningful gain ($\sim +11\%$).

Interpretation

This confirms that the channel has a **dominant angular cluster**, but it is not strictly single-path. Therefore, truncating to very few beams introduces approximation error.

3.4 Spatial Spectrum After Approximation

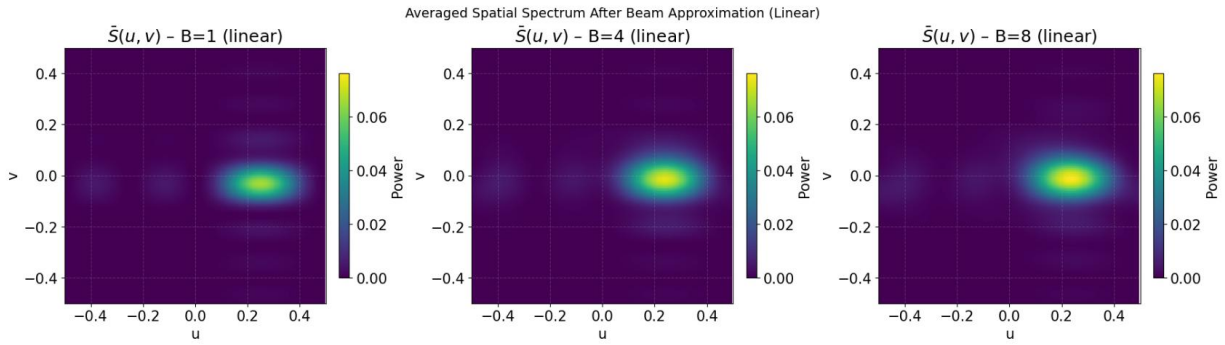


Fig 8

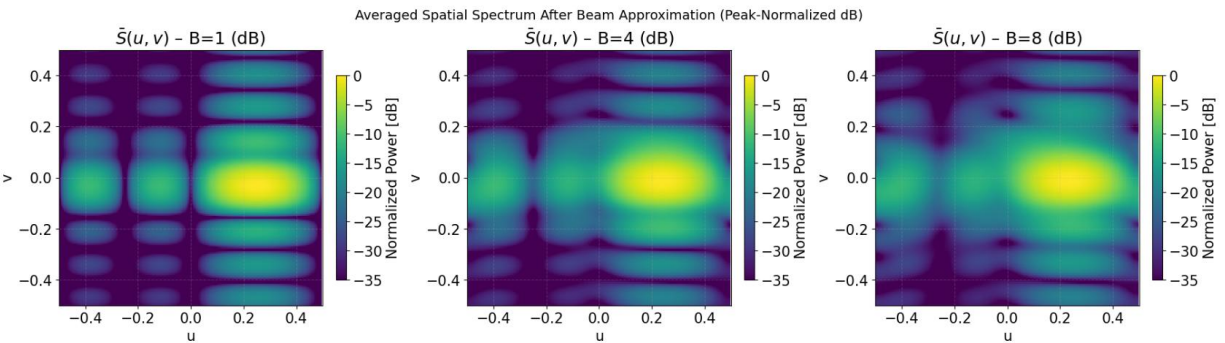


Fig 9

- **B=1:** Strong artificial side lobes and ripples appear.
- **B=4:** Ripple reduces; spectrum smoother.

- **B=8:** Closest match to original; distortion minimal.

Interpretation

Hard beam truncation causes angular distortion:

- Multiplication by a beam mask in the DFT domain produces spectral leakage.
- Smaller B results in stronger distortion and artificial ripple patterns.
- Increasing B reduces distortion and preserves spatial structure.

This explains why BER improves as B increases, the precoder becomes better aligned with the true dominant angular components.

4. Overall Conclusions

1. The channel exhibits significant angular sparsity, with nearly 47% of energy concentrated in the strongest beam.
2. However, additional beams contain non-negligible power, and ignoring them leads to performance degradation.
3. Increasing the number of retained beams improves:
 - captured channel energy,
 - precoder alignment accuracy,
 - effective SNR,
 - BER performance.
4. With B=8, more than 85% of total beam-domain energy is preserved, and performance approaches full SVD precoding.
5. Beam-domain approximation therefore provides a favorable trade-off between complexity and performance.

Control Questions:

1. Explain the concept of precoding in MIMO systems. Why is precoding essential for improving spectral efficiency and link reliability?

Precoding is a transmitter-side signal processing technique that shapes transmitted signals across multiple antennas in order to exploit channel state information (CSI).

Mathematically, the transmitted vector is:

$$\mathbf{x} = \mathbf{P}\mathbf{s}$$

where:

- $\mathbf{s}$ is the vector of transmitted data streams,
- $\mathbf{P}$ is the precoding matrix.

Precoding aligns transmitted signals with favorable spatial directions of the channel (e.g., singular vectors in SVD-based precoding).

Why precoding is essential:

- It increases spectral efficiency by enabling multiple spatial streams.
- It improves link reliability by beamforming energy toward dominant channel modes.
- It reduces inter-stream interference in multi-stream transmission.
- It maximizes received SNR by exploiting channel structure.

Thus, precoding transforms spatial diversity into either higher data rates or improved robustness.

2. What types of wireless propagation channels are known? How do different channel characteristics influence data transmission performance?

Wireless channels can be classified as:

- Line-of-Sight (LoS), direct path dominant.
- Non-Line-of-Sight (NLoS), multipath reflections dominate.
- Flat fading, constant gain over bandwidth.
- Frequency-selective fading, different gains across frequencies.
- Slow fading, channel varies slowly in time.
- Fast fading, rapid channel variation.

Channel characteristics influence performance as follows:

- Rich multipath increases spatial rank, enables multi-stream transmission.
- Strong fading reduces SNR, increases BER.
- Frequency selectivity causes inter-symbol interference.
- High mobility introduces Doppler spread.

Channel rank and scattering richness directly determine achievable spatial multiplexing gains.

3. Propose and describe an algorithm for estimating the directions of arrival (DoA). Discuss its main assumptions and limitations.

A 2D DFT-based spatial spectrum scanning algorithm was used to estimate directions of arrival. Steering vectors were constructed over a uniform angular grid, and matched-filter responses were computed across subcarriers. The averaged spatial spectrum was evaluated, and DoA estimates were obtained by locating spectrum peaks. The method assumes plane-wave

propagation and known array geometry, while resolution is limited by array aperture and grid density.

Main Algorithm:

Step 1: Channel Tensor Preparation

Given channel:

$$H \in \mathbb{C}^{(K \cdot M \cdot N)}$$

For a fixed time instant and UE antenna, extract:

$$\mathbf{h}_k \in \mathbb{C}^N$$

for each subcarrier k .

Reshape into array structure:

$$\mathbf{H}_k \rightarrow (I, P, Q)$$

where:

- $P = 4$ (horizontal dimension)
- $Q = 8$ (vertical dimension)
- $I = 2$ (polarizations)

Step 2: Construct Steering Vectors

For spatial frequencies u, v :

$$\mathbf{f}_P(u) = \frac{1}{\sqrt{P}} [1, e^{-j2\pi u}, \dots]^T$$

$$\mathbf{f}_Q(v) = \frac{1}{\sqrt{Q}} [1, e^{-j2\pi v}, \dots]^T$$

These represent hypothetical plane waves arriving from direction (u, v) .

Step 3: Compute Matched Filter Response

For each subcarrier:

$$A_1(u, v) = \mathbf{f}_P^H(u) \mathbf{H}_1 \mathbf{f}_Q(v)$$

$$A_2(u, v) = \mathbf{f}_P^H(u) \mathbf{H}_2 \mathbf{f}_Q(v)$$

Compute spatial spectrum:

$$S_k(u, v) = |A_1|^2 + |A_2|^2$$

Step 4: Average Across Subcarriers

$$\bar{S}(u, v) = \frac{1}{K} \sum_{k=1}^K S_k(u, v)$$

Step 5: Peak Detection

The estimated spatial frequency corresponds to the maximum:

$$(u^*, v^*) = \arg \max_{u, v} \bar{S}(u, v)$$

Step 6: Convert Spatial Frequency to Physical Angles

For half-wavelength spacing:

$$u = \frac{d}{\lambda} \sin(\phi) \cos(\theta)$$
$$v = \frac{d}{\lambda} \sin(\theta)$$

If $d = \lambda/2$:

$$u = \frac{1}{2} \sin(\phi) \cos(\theta)$$
$$v = \frac{1}{2} \sin(\theta)$$

Then:

$$\theta = \arcsin(2v)$$
$$\phi = \arctan\left(\frac{u}{\sqrt{1 - (2v)^2}}\right)$$

Main Assumptions

- Narrowband approximation per subcarrier
- Far-field propagation
- Plane-wave model

- Known antenna geometry
- Half-wavelength spacing
- Dominant paths produce peaks in spectrum

Limitations

- Limited angular resolution (depends on array size)
- Grid-based resolution limited by N_u, N_v
- Cannot resolve closely spaced paths if array aperture small
- Sensitive to spatial aliasing if $d > \lambda/2$
- Assumes uncorrelated multipath

4. Analyze the impact of DFT-based beam-domain channel approximation on system performance. How does the number of retained beams B affect the BER and spatial resolution?

DFT-based beam approximation retains only the strongest B angular components of the channel.

Observations from simulation:

- **Small B (e.g., 1):**
 - Large energy loss (~47% retained)
 - Significant BER degradation
 - Poor angular resolution
- **Moderate B (e.g., 4):**
 - ~74% energy retained
 - BER improves significantly
 - Main angular structure preserved
- **Larger B (e.g., 8):**
 - ~85% energy retained
 - BER close to full-channel SVD
 - High spatial resolution

Increasing B :

- Improves channel reconstruction accuracy
- Reduces mismatch between precoder and true channel
- Improves BER performance

- Increases computational complexity

Thus, beam truncation creates a trade-off between complexity and performance.

5. Explain why reliable data transmission is possible at negative signal-to-noise ratios.

The signal-to-noise ratio (SNR) is defined as

$$\text{SNR} = \frac{P_s}{P_n}$$

and in decibels:

$$\text{SNR}_{dB} = 10 \log_{10} \left(\frac{P_s}{P_n} \right).$$

A negative SNR (in dB) implies

$$P_s < P_n,$$

i.e, the average noise power exceeds the average signal power. However, reliable detection depends not only on power comparison but on the statistical separability of received signal distributions.

Single-Antenna Case

Consider BPSK transmission:

$$y = s + n,$$

where $s \in \{+1, -1\}$ and $n \sim \mathcal{N}(0, \sigma^2)$.

The conditional probability density functions are:

$$\begin{aligned} p(y | s = +1) &= \mathcal{N}(+1, \sigma^2), \\ p(y | s = -1) &= \mathcal{N}(-1, \sigma^2). \end{aligned}$$

The optimal detector decides based on the sign of y . The bit error probability is

$$P_e = Q \left(\sqrt{\frac{2E_b}{N_0}} \right),$$

where

$$\frac{E_b}{N_0} = \text{SNR}.$$

Even when $\text{SNR}_{dB} < 0$, the argument of the Q-function remains positive, meaning

$$P_e < 0.5.$$

Thus, detection performance is still better than random guessing as long as distributions do not completely overlap.

MIMO with SVD Precoding

For a MIMO system:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}.$$

With SVD decomposition:

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H,$$

and single-stream precoding:

$$\mathbf{x} = \mathbf{v}_1 s,$$

the received signal becomes

$$\mathbf{y} = \sigma_1 \mathbf{u}_1 s + \mathbf{n}.$$

After matched filtering with $\mathbf{u}_1^H$:

$$\hat{s} = \sigma_1 s + \tilde{n}.$$

The effective SNR is therefore

$$\text{SNR}_{\text{eff}} = \sigma_1^2 \cdot \text{SNR}.$$

If the dominant singular value σ_1 is sufficiently large, the effective SNR may become positive even when the raw SNR is negative. This spatial gain explains why BER can remain low in beamformed MIMO systems.

Multi-Antenna Combining Gain

With M receive antennas and maximal ratio combining:

$$\text{SNR}_{\text{eff}} = \left(\sum_{i=1}^M |h_i|^2 \right) \text{SNR}.$$

Thus, diversity and array gain increase effective SNR proportionally to channel strength.

Conclusion

Although negative SNR implies that noise power exceeds signal power, reliable transmission remains possible because:

1. Detection depends on probability distributions rather than instantaneous amplitudes.
2. Error probability follows the Q-function, which remains below 0.5 for finite SNR.
3. Spatial processing (beamforming) increases effective SNR via dominant singular values.
4. Diversity combining further enhances reliability.

Therefore, even at negative SNR values, the bit error rate can remain significantly below 0.5, enabling reliable communication.

SNR (dB)	Linear SNR	BER (BPSK approx)
0 dB	1	~0.078
-3 dB	0.5	~0.16
-6 dB	0.25	~0.24
-10 dB	0.1	~0.37
$-\infty$ dB	0	0.5

6. What is the physical meaning of spatial frequency in antenna arrays? How is it related to the direction of arrival?

Spatial frequency describes how the phase of an incoming electromagnetic wave changes across the elements of an antenna array. When a plane wave arrives from a certain direction, it reaches each antenna element with a different propagation delay, which results in a progressive phase shift between adjacent antennas.

For an array with inter-element spacing d and wavelength λ , the phase shift between neighboring antennas is

$$\Delta\varphi = \frac{2\pi d}{\lambda} \sin(\theta),$$

where θ is the angle of arrival.

The normalized spatial frequency is defined as

$$u = \frac{d}{\lambda} \sin(\theta).$$

Thus, spatial frequency represents the rate of phase rotation across the array. Larger angles of arrival produce larger spatial frequencies, meaning faster phase variation between antennas.

In planar arrays, two spatial frequencies (u, v) are used to describe phase variation along horizontal and vertical dimensions. Each pair (u, v) corresponds to a specific physical direction of arrival (ϕ, θ).

Therefore, spatial frequency is a mathematical representation of the wave's direction, and direction-of-arrival estimation is performed by identifying peaks in the spatial spectrum over the (u, v) domain.

7. Explain the phenomenon of spatial aliasing and grating lobes. Under what conditions do they occur?

Spatial aliasing is a phenomenon in antenna arrays that occurs when the array elements are spaced too far apart relative to the signal wavelength. It is the spatial equivalent of temporal aliasing in signal sampling.

For a uniform linear array with inter-element spacing d , the spatial frequency is

$$u = \frac{d}{\lambda} \sin(\theta),$$

where λ is the wavelength and θ is the angle of arrival.

To ensure a unique mapping between spatial frequency and physical angle, the sampling condition must satisfy

$$d \leq \frac{\lambda}{2}.$$

If

$$d > \frac{\lambda}{2},$$

multiple angles produce the same spatial frequency. As a result, the array response exhibits additional undesired maxima called grating lobes.

Grating lobes are secondary peaks in the radiation pattern that have comparable amplitude to the main lobe. They create ambiguity in direction-of-arrival estimation because the array cannot distinguish between the true direction and false directions corresponding to these lobes.

Therefore, spatial aliasing and grating lobes occur when the antenna spacing exceeds half the wavelength, violating the spatial sampling theorem. To avoid these effects, antenna arrays are typically designed with half-wavelength spacing.

8. Explain the principle of water-filling power allocation in multi-stream MIMO systems.

In multi-stream MIMO systems, the channel matrix can be decomposed using singular value decomposition (SVD):

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H,$$

where $\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ contains the singular values. After SVD precoding and combining, the MIMO channel is transformed into r parallel independent subchannels:

$$y_i = \sigma_i s_i + n_i.$$

If power P_i is allocated to the i -th stream, the effective SNR becomes

$$\text{SNR}_i = \frac{P_i \sigma_i^2}{N_0}.$$

The total channel capacity is therefore

$$C = \sum_{i=1}^r \log_2 \left(1 + \frac{P_i \sigma_i^2}{N_0} \right),$$

subject to the total transmit power constraint

$$\sum_{i=1}^r P_i = P_{\text{total}}.$$

To maximize capacity under this constraint, the optimal power allocation is given by the water-filling solution:

$$P_i = \max \left(0, \mu - \frac{N_0}{\sigma_i^2} \right),$$

where μ is chosen such that the total power constraint is satisfied.

This solution allocates more power to stronger channel modes (large σ_i) and may allocate zero power to weak modes. The term “water-filling” arises from the analogy of pouring water into containers with different base heights: better channels (lower $\frac{N_0}{\sigma_i^2}$) receive more power, while poor channels may remain unused. This strategy maximizes spectral efficiency and overall system capacity.

9. Why is BER an appropriate performance metric in digital communication systems? What are its limitations?

The Bit Error Rate (BER) is defined as

$$\text{BER} = \frac{\text{Number of incorrectly detected bits}}{\text{Total number of transmitted bits}}.$$

BER directly measures the reliability of a digital communication system by quantifying how often transmitted bits are received incorrectly. It is widely used because:

- It reflects the actual performance of modulation, precoding, and detection algorithms.
- It allows direct comparison between different transmission schemes.
- It depends on signal-to-noise ratio (SNR), channel conditions, and processing techniques.
- It is simple to compute in simulations and experiments.

Since digital communication systems ultimately transmit binary data, BER provides a fundamental and intuitive measure of link quality.

Limitations of BER

Despite its usefulness, BER has several limitations:

1. **Does not measure throughput:** A system with low BER may still have low data rate.
2. **Does not reflect spectral efficiency:** Higher-order modulation may increase throughput but also increase BER.
3. **Ignores latency and delay:** BER does not account for transmission delays or real-time constraints.

4. **Does not consider coding overhead:** Channel coding reduces BER but also reduces effective data rate.
5. **Not always suitable for packet-based systems:** In practical systems, Block Error Rate (BLER) or Packet Error Rate (PER) may be more relevant.

Conclusion

BER is an appropriate performance metric because it directly measures transmission reliability at the bit level. However, it does not fully characterize system performance and therefore should be considered together with other metrics such as spectral efficiency, throughput, and block error rate.

10. What other performance metric we can in digital communication systems?

In addition to Bit Error Rate (BER), several other performance metrics are commonly used to evaluate digital communication systems:

1. Spectral Efficiency

$$\eta = \frac{R}{B} [\text{bits/s/Hz}]$$

where R is the data rate and B is the bandwidth.

It measures how efficiently the available spectrum is utilized.

2. Channel Capacity

$$C = B \log_2(1 + \text{SNR})$$

It represents the theoretical maximum achievable data rate under given channel conditions.

3. Throughput

The actual successfully delivered data rate, considering retransmissions and protocol overhead.

4. Block Error Rate (BLER) / Packet Error Rate (PER)

These measure the probability that an entire block or packet is received incorrectly, which is often more relevant in packet-based systems.

5. Signal-to-Noise Ratio (SNR)

These quantify link quality and directly influence achievable performance.

6. Energy Efficiency

$$\text{Energy Efficiency} = \frac{\text{bits}}{\text{Joule}}$$

7. Error Vector Magnitude (EVM)

Measures modulation quality by quantifying deviation of received constellation points from ideal positions.

Conclusion

While BER measures transmission reliability at the bit level, a complete system evaluation typically requires additional metrics such as spectral efficiency, capacity, throughput, and block error rate to capture efficiency, robustness, and practical system performance.

11. What other important processing blocks were omitted in our simulation code? What are their role in real communication systems?

The simulation focused primarily on spatial processing (SVD precoding, beam-domain approximation, and equalization). However, several important processing blocks commonly used in real communication systems were not included.

1. **Channel Coding:** Techniques such as LDPC, Turbo, or Polar codes add redundancy to transmitted data to enable error correction at the receiver. Coding significantly reduces BER and allows reliable transmission at lower SNR values.
2. **Channel Estimation**
In practice, the channel matrix $\mathbf{H}$ is not perfectly known and must be estimated using pilot signals. Estimation errors degrade system performance.
3. **Synchronization (Time and Frequency)**
Real systems require accurate timing synchronization and carrier frequency offset (CFO) compensation. Without synchronization, symbol detection becomes unreliable.
4. **Power Control and Adaptive Modulation**
Transmission parameters (modulation order, coding rate, power allocation) are dynamically adjusted based on channel conditions to maximize throughput while maintaining reliability.
5. **Hybrid Beamforming Constraints**
Practical systems may use analog-digital hybrid beamforming, where the precoding matrix is constrained by hardware limitations.
6. **RF Impairments and Hardware Effects**
Nonlinear power amplifiers, phase noise, quantization errors, and hardware imperfections affect real system performance but were not modeled.

Conclusion

The simulation modeled an idealized MIMO system with perfect channel knowledge and no hardware impairments. In real communication systems, additional processing blocks such as channel coding, synchronization, estimation, and adaptive control are essential to ensure reliable and efficient data transmission under practical conditions.